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Automatic Detection Pipeline for Accessing the Motor Severity of Parkinson's Disease in Finger Tapping and Postural Stability

NING YANG[1,2], DE-FENG LIU[3], TAO LIU[4,5], TIANYUAN HAN[4,5], PINGYUE ZHANG[4,5], XUENAN XU[4,5], SIYU LOU[4,5], HUAN-GUANG LIU[3,6,7], AN-CHAO YANG[3,6,7], CHENG DONG[8,9], MANG I. VAI[1,2,10], (Senior Member, IEEE), SIO HANG PUN[2], AND JIAN-GUO ZHANG[3,6,7]

[1]Department of Electrical and Computer Engineering, Faculty of Science and Technology, University of Macau, Taipa, Macau 999078, China
[2]State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau, Taipa, Macau 999078, China
[3]Department of Neurosurgery, Beijing Tiantan Hospital, Capital Medical University, Beijing 100050, China
[4]MoE Key Laboratory of Artificial Intelligence, AI Institute, Shanghai Jiao Tong University, Shanghai 201100, China
[5]X-LANCE Laboratory, AI Institute, Shanghai Jiao Tong University, Shanghai 201100, China
[6]Department of Functional Neurosurgery, Beijing Neurosurgical Institute, Capital Medical University, Beijing 100050, China
[7]Beijing Key Laboratory of Neurostimulation, Beijing 100050, China
[8]School of Intelligent Systems Science and Engineering, Jinan University, Zhuhai 519070, China
[9]Biomedical Engineering Postdoctoral Research Station, School of Life Science and Technology, Jinan University, Guangzhou 510632, China
[10]Key Laboratory of Medical Instrumentation and Pharmaceutical Technology of Fujian Province, Fuzhou 350116, China

Corresponding authors: Sio Hang Pun (lodgepun@um.edu.mo) and Jian-Guo Zhang (zjguo73@126.com)

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ABSTRACT Parkinson's disease (PD) is a nervous disorder that can cause motor impairment. PD severity assessment based on a series of motor movements illustrated in Unified Parkinson's Disease Rating Scale (UPDRS) is an important part of clinical PD diagnosis. However, the current quantifying method heavily relies on human observation, which is time-consuming and subjective. Therefore, automatic severity estimation stemming from machine learning methods is receiving an increasing amount of research attention. However, these advances are still limited by data availability and interpretability. In this paper, we release a large PD motor dataset of over 300 real PD patients collected under doctors' instructions and propose a pipeline to automatically quantify the motor severity of PD in finger tapping and postural stability. These two selected movements are representative of local and global motor control, exhibiting great clinical importance. The pipeline contains three-stage: pose estimation, domain knowledge extraction, and classification stage. The pose estimation uses deep-learning-based methods to extract 21 and 17 key points for finger tapping and postural stability respectively. The domain knowledge extraction stage extracts several explicit features pre-defined by experienced neuro-physicians. Finally, a classifier is trained to infer PD severity under MDS-UPDRS. To combine deep-learning-based features from pose estimation and domain features from the expert, the pipeline achieves a better trade-off between the model efficiency and clinical interpretability. Experiments show that our method achieves a micro average f1-score of 88%, 84%, and 84%, respectively on left finger tapping, right finger tapping, and postural stability, outperforming previous methods by a large margin. In addition, involving expert knowledge in the feature extraction stage greatly improves our model's interpretability, which is essential in automatic PD detection.

INDEX TERMS Activity recognition, deep learning, finger tapping, neural network, Parkinson's disease, pose estimation, postural stability.

I. INTRODUCTION

Parkinson's disease (PD) is a nervous disorder, which often occurs in the elderly over the age of 60. Millions of people

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worldwide have been diagnosed with this disease. PD is often combined with motor impairment (MI) such as rigidity, tremor, or slowness in any body parts of a patient. An important part of the diagnosis of PD is assessing PD severity objectively and timely. Patients are instructed to perform specific motor actions, followed by professionals' severity examination based on those actions. In practice, PD severity is often observed manually by doctors. The manual assessment is time-consuming and the result may vary among doctors due to their subjectiveness or training experiences. Research on automatic PD evaluation via motor movement has drawn an increasing amount of attention. For example, one approach to evaluate PD severity is based on wearable sensors [1]–[8]. Nevertheless, sensor-based approaches require physical devices and can rarely be used in remote places or online. With the rapid development of deep learning, multiple computer vision tasks like pose estimation [9] and action recognition [10], [11] oversee a performance boost. Combined with these works, vision-based methods [12]–[21] has also witnessed successful application on detecting PD motor severity.

However, those deep learning algorithms work in an end-to-end format, extracting features automatically, leading to low interpretability and explainability, which is essential for such clinical-driven applications.

In this paper, we propose a deep-learning-based PD detection method with knowledge-infused PD-specific features to partially alleviate the aforementioned opacity problem, further combining the model efficiency and interpretability altogether. Specifically, we collected motor function video recording data of over 300 real PD patients' finger tapping and postural stability experiment. Experienced clinicians labeled PD severity degree and helped define a novel PD-specific feature-set for these two actions. The newly proposed featureset is in line with MDS-UPDRS, while providing precise and quantitative descriptions of the severity scaling hence further improving the inter-rater agreement. On such a basis, we achieve better PD detection performance compared with the traditional general feature extraction method. Since the newly proposed feature-set is defined under stringent clinical supervision, helpful interpretations can be drawn from our model's predicted output.

Detecting PD via a set of motor movement is vastly used in clinical settings. The most broadly-used assessment procedure is MDS-UPDRS [22], a clinimetric assessment of the Movement Disorder Society (MDS)-sponsored revision of the Unified Parkinson's Disease Rating Scale. MDS-UPDRS contains four parts: non-motor experiences of daily living, motor experiences of daily living, motor examination, and motor complications. We focus on the MDS-UPDRSIII exercises: motor examination. The motor examination part has 18 examination items and we choose two parts which are finger tapping and postural stability, corresponding to part 3.4 and part 3.12 in MDS-UPDRS respectively. Finger tapping refers to the PD patient tapping the index finger on the thumb as quickly as they can. The purpose of finger tapping testing is to measure the patient's ability to control their hand. Postural stability detection refers to how well the patient keeps stability while the doctor pulls the shoulder of the patient from the back. In our paper, we focus on these actions as they respectively represent the local posture and overall posture of the PD patient. The output for each action is a 5-scale severity measurement: normal, slight, mild, moderate, and severe. The disease's nature decides that the distribution of these five categories is imbalanced: severe patients are in particular rare as such a group is usually incapacitated.

In an automatic PD detection framework, feature input is always crucial to the final detection performance. Previous work has covered various feature-sets including kinematic, spectral features, etc. For instance, Li *et al.* [14] define 32 features to train a classification or regression task. Those features are 15 kinematic features, 16 spectral features, and one is the convex hull. Goyal *et al.* [16] propose five features to detect the type of neurological disorder in the gait action. Five features are limb straightness, hand-leg coordination, upper body straightness, body straightness, and mutual distances. However, these features are either action-universal hence lacking specificity, or only targeted at gaiting, overlooking other commonly-adopted clinical assessing motor movement. Inspired by the above methods and following MDS-UPDRS, we newly propose knowledge-infused PD-specific featureset for the finger tapping action and postural stability actions. For the finger tapping, we propose three features: tapping rate, tapping frozen time, and tapping amplitude. For postural stability, five features are identified: upper-body straightness, body straightness, lower-body straightness, backward steps, and supporting status.

Another vital challenge in automatic PD detection lies in data scarcity. By far, the PD dataset in [13], [16], [23] is simulated, not publically available, or in a small dataset scale. So we collect a PD motor dataset from real scenarios for finger tapping and postural stability. All data is collected from a top-class hospital specialized in PD assessment and treatment. Real PD patients participated in the video recording and completed the two aforementioned motor actions guided by professional examiners. Experienced doctors have been invited to label the severity of their PD symptoms by adopting MDS-UPDRS for each patient. The dataset will be released to the community.

The main contribution of this work is threefold as below.

1. A deep-learning-based automatic PD motor severity detection method in finger tapping and postural stability that are well-performing in PD severity prediction while of high interpretability. This pipeline has three stages: the pose extraction stage, domain knowledge extraction stage, and classification stage. The final output of our pipeline is the severity of MDS-UPDRS. The pose extraction and the classification stage use deep learning-based methods to guarantee performance.
2. We propose a clinician-guided feature-set specifically tailored for PD detection on the basis of a local and global action to assess a PD patient's motor function

based on MDS-UPDRS. Three features for the finger tapping action and five features for the postural stability. The tapping rate, the tapping frozen times, and the tapping amplitude variation are extracted for the finger tapping action. Upper-body straightness, body straightness, lower-body straightness, backward steps, and supporting status are extracted for postural stability. Those features add interpretability to our pipeline.

3. We collect a large PD motor dataset from the hospital with 368, 243, and 168 real PD patients for left finger tapping, right finger tapping, and postural stability. We release the dataset and provide a testbed to the community.

II. RELATED WORKS

Methods to access the severity of PD patients can be categorized as the sensor-based or the vision-based method. The sensor-based methods [1]–[8] use wearable sensors like accelerometer and gyroscope to measure the continuous sensor signal of the body movement. Then several machine learning approaches are trained to classify those signals into different classes of PD severity. The vision-based methods [13]–[16], [20] refer to detecting the severity of PD patients via some image processing methods such as human action recognition. Although the sensor-based methods are sensitive to body action, those methods require patients to wear different sensors, which limits their wide application. In our paper, we adopt the vision-based method.

Vision-based human action recognition [10] is to infer human actions from videos. In PD assessment, the action of PD patients is the inferred content. The input type of the video frame has two typical types: image data, and skeleton data. Image data mainly refers to RGB images or the optical flow. C3D [24], I3D [25], [26], TSN [27] and SlowFast [28] are the typical methods. Skeleton data refers to key points extracted by pose estimation [9]. Human action recognition with skeleton input is also called skeleton-based action recognition. The method takes human key points as the input. So the method usually consists of two stages: pose estimation stage and classifying stage. The former stage extracts human key points and the latter is to classify the action. PoseC3D [29] is the typical method. In our experiment, we take I3D [25], [26] and PoseC3D [29] as the baseline.

Various action recognition methods are conducted in PD assessment. Dadashzadeh *et al.* [13] directly take videos of PD actions as the input of the model and uses I3D [25], [26] algorithm to measure the PD severity in the hand movement and the gait. They test on their proposed dataset and obtain 72.3% and 77.1% top-1 accuracy on hand movement and gait tasks respectively. Different from taking raw videos as the input, Li *et al.* [14] use a pose estimation, CPM [30], to extract 13 key points before accessing the PD severity of leg agility and toe-tapping. Mehrizi *et al.* [15] use the output of 3D body pose to classify whether the person has PD. Goyal *et al.* [16] use the result of pose estimation to access the gait abnormality of PD. They achieve a test accuracy of 86.4% in their gait dataset. Hao Li *et al.* [31] propose a three-stage classification network to measure the PD severity. The input of the network is the skeleton data and three-stream architecture grasps the pose, motion, and geometric features in a unified way. Zhilin Guo *et al.* [32] adopt a 3D vision-based method to evaluate PD hand severity 3D hand pose and depth camera.

As a widely used method in PD motor severity assessment and the part of the skeleton-based human action recognition, pose estimation [9] is the algorithm to estimate key points of the whole body or part of the body e.g., the hand. Estimating the key point of the whole body is defined as the body pose estimation and the hand's is defined as the hand pose estimation. The location of key points is representative of human action. In PD motor severity assessment, affected by action rigidity, tremor, or slowness, the action of the human pose for the PD patient can reflect different motor severities. For the dataset of pose estimation, the number of whole-body joints [33]–[35] ranges from 10 to 17. For example, MS COCO dataset [34] contains 17 key points: the nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles. The number of hand joints [36]–[41] ranges from 14 to 21. For example, OneHand10K [41] dataset contains 21 key points including one key point for the wrist and four key points for each hand. Those datasets are collected in the wild with various gestures. So we adopt those datasets to train the pose estimation method.

Pose estimation can be further classified as the top-down or the bottom-up method. The top-down method [42], [43] requires detecting the bounding boxes of human or hand first and then extracting pose for each bounding box, while the bottom-up method first predicts all joints for every human or hand and group them with algorithms [42], [43]. Bottom-up methods for human pose estimation will deteriorate when multiple persons overlap. In our dataset, postural stability contains the interaction between the patient and the examiner. The detection of postural stability action will also use the bottom-up method. Because there is only one hand in the frame at the same time, hand pose estimation can use either top-down or bottom-up methods. We also use top-down methods for simplicity.

PD motor dataset is a crucial part of testing the algorithm of PD motor severity detection. Goyal *et al.* [16] propose a video dataset called NeuroSynGait to detect gait abnormalities. This dataset contains 258 videos and all videos are labeled with five classes. But this dataset is simulated by healthy humans and not collected with real patients. Dadashzadeh *et al.* [13] collect finger tapping and gait from 25 PD patients several times during 8-week intervals. They have collected 1058 videos in total. Lu *et al.* [13] collected gait from 30 PD patients. To augment the normal type, they add 20 non-PD patients from CASIA Gait Database A [23]. Since collecting large amounts of PD patients is time-consuming and challenging, those collected datasets [13], [23] are on a small scale, which hinders the development of methods to detect the PD motor severity. In this paper, we propose a PD motor dataset from real scenarios in two actions: finger tapping and postural stability. We adopt this dataset to test our method.

(a) Finger Tapping (b) Postural Stability

FIGURE 1. Parkinson's Disease Motor Dataset. We collect a PD motor dataset with two actions: (a) Finger tapping: the patient taps the index finger on the thumb as fast as they can. (b) Postural stability: the examiner pulls the shoulder of the patient from the back to test their ability to keep stability. Our dataset is collected from the hospital with real patients.

TABLE 1. PD motor dataset metrics. #videos: The number of videos. frame ratio: The minimum/median/maximum ratio of video height to width. length (s): The minimum/average/maximum video length in seconds. #types: The type distribution of PD severity in numbers. The PD severity type of the finger tapping is normal, slight, mild, and moderate of MDS-UPDRS respectively. The PD severity type of the postural stability is normal, slight, and moderate of MDS-UPDRS respectively. Several severity types are missing due to limitations in the patient and the environment.P

Type	Finger Tapping (Left)	Finger Tapping (Right)	Postural Stability
#videos	368	243	168
frame ratio	0.96 / 1.39 / 2.39	0.95 / 1.40 / 2.20	0.56 / 0.56 / 0.56
length (s)	5.00 / 5.00 / 5.00	5.00 / 5.00 / 5.00	1.00 / 3.86 / 14.60
#types	66 / 134 / 141 / 27	33 / 113 / 88 / 9	115 / 15 / 38

III. DATASET

We propose a PD motor dataset.[1] The dataset contains two motor actions: finger tapping, and postural stability. Those two actions correspond to part 3.4, and part 3.12 in MDS-UPDRS respectively. Finger tapping contains both left and right hand. Finger tapping requires the patients to tap their thumb and index finger as fast as they can. Postural stability is an interaction action. One examiner, usually the doctor, pulls the shoulder of the patient and the patient tries their best to keep the stability. The reasons to choose those two actions are that finger tapping and postural stability represent local and global motor impairment respectively.

Videos in our dataset are collected from Beijing Tiantan Hospital, a Chinese hospital, with 368, 243, and 168 real PD patients for left finger tapping, right finger tapping, and postural stability, respectively. The numbers are different because some patients are not recorded all the actions. Of those recorded patients, 53% are males, and the rest are females. The age of PD patients is from 41 to 79, and the average is 63. The onset time of those patients ranges from 1 to 27 years, and the average onset time is nine years. This dataset is approved by the Ethics Committee of Beijing Tiantan Hospital.[2]

Following the MDS-UPDRS, each action is labeled from 0 to 4, representing normal, slight, mild, moderate, and severe types, respectively. Same as reference [13], our dataset also lacks severe degree, which is subject to conditions that those patients usually can not walk or need the assistance of others. So labels with the severe type are all excluded in our dataset. Besides, the postural stability lacks a mild type because the recording environment was in the corridor, lacking enough room for patients to step back.

[1] The dataset is available at https://github.com/pddata/PDMotorDB.
[2] No. KY 2018-008-02.

TABLE 2. The Kapper coefficient for three labeled doctors. Values in each grid represent the Kapper coefficient for paired doctors in the left finger tapping, right finger tapping, and postural stability, respectively.

	Doctor #1	Doctor #2	Doctor #3
Doctor #1	n/a	0.44 / 0.66 / 0.63	0.34 / 0.28 / 0.84
Doctor #2	0.44 / 0.66 / 0.63	n/a	0.38 / 0.36 / 0.68
Doctor #3	0.34 / 0.28 / 0.84	0.38 / 0.36 / 0.68	n/a

We invite three experienced doctors for labeling and calculate the Kappa coefficient for them. The result is shown in Table 2. We calculate the Kappa coefficient for each paired doctor and the grid in the table has three numbers, representing left finger tapping, right finger tapping, and postural stability, respectively. The higher the number is, the better consistency the doctors have. We do not measure the intra-rater coefficient here, which is marked as n/a in the table. To make the label more reliable and definite, we also do the following criteria: (a) If two or more doctors agree with the label, the agreed label is assigned to the instance. (b) If all three doctors disagree, we take the average and assign the result to the closest label. For example, if the labels of three doctors are 0, 1, and 2 and the average is 1, label 1 will be assigned to the instance. After that, we will get a unique label for all instances.

All actions are recorded in videos by *SONY-FDR-AX100E*, an RGB camera. The video frame rate is set to 25 fps with a resolution of 1920×1080. The patients stand or sit in front of the camera. Then, we manually crop or split those actions from raw recorded videos. Our released dataset contains raw videos except for face parts blurred out to protect the patient's privacy. To better interpret the collecting criteria, we will describe each action in detail.

A. FINGER TAPPING

From the 3.4 part of MDS-UPDRS, there exist several criteria, e.g., the interruption occurring, the tapping rate being slower, or the amplitude decreasing in finger tapping, which is used for doctors measuring the severity type. In practice, patients sometimes can not tap in an accurate tapping count, ten times. Besides, the average clinician Kappa coefficient in finger tapping for both hands is the 0.13, a slight consistency. If we adopt the original criteria in UPDRS, the low consistency affects the model stability. So, in the finger tapping part, the clinician slightly adjusted the rule of MDS-UPDRS's 3.4 part from 10 times to 5 seconds. Besides, the criteria of the corresponding label are also modified. For example, the mild type is modified to the following criteria: a) 30% to 50% interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 5 seconds tap sequence. Meeting any one of the criteria will be judged the mild type. After those criteria, we calculate the average Kappa, which increases from the 0.13 to the 0.41, representing the Kappa consistency improving from the slight to the moderate.

After the recording, videos with finger tapping are manually cropped to ensure that the hands of the patients are the central part of the video frame. The video length in the finger

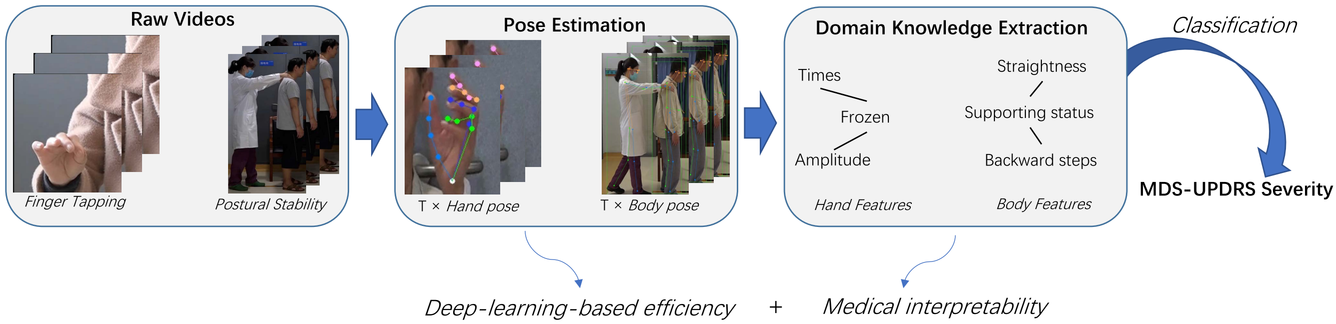


FIGURE 2. **Automatic detection pipeline for PD motor severity in the finger tapping and postural stability. Our pipeline first uses pose estimation, based on deep-learning methods, to extract 17 and 21 key points for body and hand respectively. The deep-learning methods show the efficiency in extracting those key points. However, the deep-learning methods lack interpretability and require a large amount of data. To avoid the shortcomings, we extract knowledge-infused PD-specific features for each action. Those features are manually designed and selected by the expert. In our experiment, we extract three hand domain features including the tapping rate, the tapping frozen times, and the tapping amplitude variation for the finger tapping action, and five body domain features including upper-body straightness, body straightness, lower-body straightness, backward steps, and supporting status. Those features reflect the PD patient's ability to control their hands or body. After extracting those domain features, a classification method, either machine learning methods or deep-learning-based methods, is adopted to make the final classification. Through this pipeline, we can get a well trade-off between deep-learning efficiency and medical interpretability.**

tapping action is fixed to five seconds long, while the video resolutions are various, which are affected by the various hand distance to the camera. The median ratio of height to width is 1.39 and 1.4 for left and right hand, respectively. Because patients with the severe class hardly tap their finger, the finger tapping only has the normal, slight, mild, and moderate PD severity of MDS-UPDRS and lacks the severe type.

B. POSTURAL STABILITY

Postural stability detection refers to how well a patient keeps stability while the doctor pulls the shoulder of the patient from the back. Following the instructions of part 3.12 in the MDS-UPDRS, the doctor stands behind the back of the patient at least 1-2 meters away and pulls the shoulder of the patient to test. The patient is allowed to take a step backward to avoid falling. At first, the doctor will try one or two times to test whether the response of the patient is correct. After the patient fully understands the instruction, the doctor pulls the shoulder of the patient briskly and forcefully.

In order to fully observe the status of the patient, a camera is on the side of the patient and the doctor, as shown in Figure 1(b). Only final pulling is recorded. The postural stability action does not require cropping and is saved with the original resolution of 1920×1080. However, the action time varies due to the time for the patients to keep stability varying. The action time in postural stability ranges from 1 to 14.6 seconds and the average and the median are 3.86 and 3 seconds respectively. As is denoted in Section III, our data set lacks the mild type in this part. Besides, due to patients with the severe type hardly standing, our data set also lacks the severe type in this part.

IV. METHOD

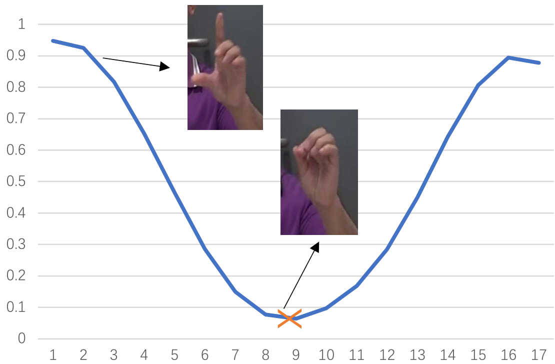
Our pipeline has three stages: pose estimation stage, domain knowledge extraction stage, and a classification stage. In the pose estimation stage, we adopt pose estimation to extract key points for body and hand respectively. In the domain knowledge extraction stage, we propose several domain features for each action. In the final classification stage, we train a classifier to get the severity type of different actions. The detailed pipeline is shown in Figure 2.

A. FINGER TAPPING DETECTION

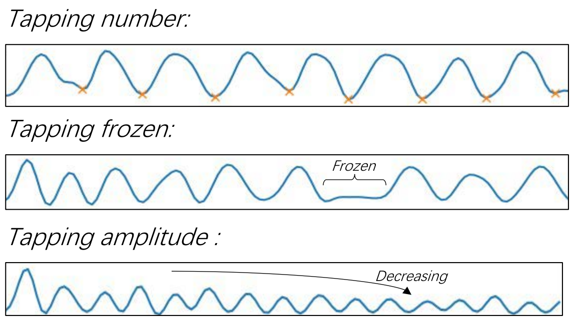
In the pose estimation stage, we extract 21 hand key points via hand pose estimation. 21 key points consist of four key points in the index finger, four key points in the middle finger, four key points in the ring finger, four key points in the pinky finger, and a thumb key point. From the part 3.4 of MDS-UPDRS [22], the distance from the thumb and the index finger is a crucial criterion to judge the severity of finger tapping. So we use two key points in the following stage: one is the thumb key point and the other is the furthest point from the palm heel in the index finger.

In the domain knowledge extraction stage, we extract three features: the tapping rate, the tapping frozen times, and the tapping amplitude variation for the finger tapping action. Those features represent the patient's ability to control their fingers. The tapping rate is defined as the tapping number divided by the time interval. Tapping frozen times is defined as the hesitation number of the patient. The tapping amplitude variation refers to the distance variation in which the patient spread the thumb and the index finger.

To calculate the three features above, we first calculate the distance from the thumb to the index finger for each frame. Finger tapping status can be represented by the distances. Those distances compose a discrete signal, which can form a wave form by connecting the adjacent point. To make the wave form smooth and filter noise, we use low pass filtering to get rid of the frequency of over 4 Hz. Second, we normalize the distance by dividing by the maximum distance to avoid the variation of the hand distance to the camera. For example, when the distance comes to zero, the thumb and the index



(a) Wave form of the finger tapping



(b) Examples of finger tapping feature

FIGURE 3. Domain features of finger tapping (a) Wave form of the finger tapping: the distances from the thumb to the index finger varies with time, which forms a digital signal wave form. The vertical axis is normalized. The horizontal axis is time and the scale represents 40 ms. (b) Examples of finger tapping feature: 1) Tapping action can be represented by the local minima of the wave form and the tapping number can be calculated by the number of the local minima. The mark *'x'* is the local minima point in this wave form; 2) The tapping frozen is the patient keeping a still action. There is a flat line in this wave form; 3) The tapping amplitude is the amplitude variation with time. There exists a decrease in the amplitude in this wave form.

finger are pressed together. When the distance comes to one, the thumb and the index finger are pulled apart at most. The wave form of one cycle can be viewed as a tapping, shown in Figure 3(a). The mark *'x'* in yellow represents the local minimal point.

From the wave form, the tapping rate can be calculated as the tapping number divided by time. For example, in the first row of Figure 3(b), the tapping rate is the number of *'x'* mark, the local minimal point divided by five seconds. The tapping rate is 1.6 taps per second in this example. The frozen state is defined as the patient prolonging the current action for a while. In the second row of Figure 3(b), we can see there is an obvious flat line in the center of the wave form. We first calculate the first derivative of the wave form and take the range with small derivatives as the frozen interval. We take the number of the frozen interval as the feature of tapping frozen times. The tapping amplitude refers to the patient's ability to keep the tapping amplitude. For example, there is an obvious decrease in the amplitude of the wave form, shown in the third row of Figure 3(b). We calculate the standard deviation of the tapping amplitude to grasp the tapping amplitude variation,

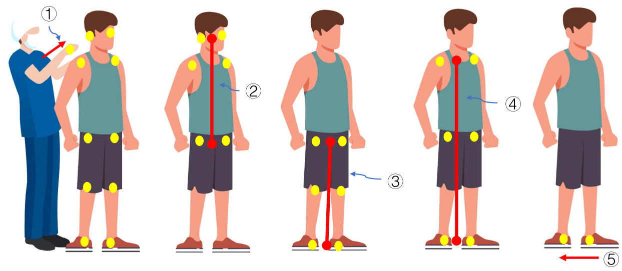


FIGURE 4. Key points used in extracting the features of the postural stability. Ten key points of the patient are two ear key points, two shoulder key points, two hip key points, two knee key points, and two ankle key points. Two points of the examiner are the left hand and the right hand. Number one to five in the figure represent features for supporting status, upper-body straightness (US), lower-body straightness (LS), body straightness (BS), and backward step, respectively.

avoiding the effects of the various absolute pixel distances. Besides, this is also consistent with the criteria in the 3.12 part of MDS-UPDRS.

In the final classification stage, three features represent three dimensions. We feed the data with three dimensions to train a classifier, and any classification algorithms can be adopted here. We adopt the DNN model to make the classification and also conduct several machine learning methods in our ablation studies. In our finger tapping data set, we classify four types: normal, slight, mild, and moderate type of MDS-UPDRS.

B. POSTURAL STABILITY DETECTION

In the pose estimation stage, we first extract the 17 body key points for the patient and the examiner respectively in each video frame. 17 key points contain key points of the nose, eye, ear, shoulder, elbow, wrist, hip, knee, and ankle. The nose only has one key point and the rest have two key points respectively. Ten key points of the patient are used in the next stage and those key points are two ear key points, two shoulder key points, two hip key points, two knee key points, and two ankle key points. Only two points, the left hand and the right hand, of the examiner are used. See Figure 4 for details.

In the domain knowledge extraction stage, we form five domain-knowledge-infused features: upper-body straightness, body straightness, lower-body straightness, backward steps, and supporting status. Upper-body straightness and body straightness are same as reference [16]. Upper-body straightness is the offset from the mid-point of the shoulders to the line from the hips to the ears. Body straightness is the offset from the mid-point of the hips to the line from the ankles to the shoulders. Similar to the upper-body and body straightness, we also define the lower-body straightness which is the offset from the mid-point of knees to the line from the hips to the ankles. Upper-body straightness, body straightness, and lower-body straightness can be calculated in Equation. (1), Equation. (2) and Equation. (3), as shown at the bottom of the next page. We use 2D pose key point here.

TABLE 3. Notations for the used key points in the postural stability. Corresponding to 2D pose estimation, each key point has *x* and *y* coordinates.

Used key points	Notation	Belonging
Left Ear	$E_l(x), E_l(y)$	Patient
Right Ear	$E_r(x), E_r(y)$	Patient
Left Shoulder	$S_l(x), S_l(y)$	Patient
Right Shoulder	$S_r(x), S_r(y)$	Patient
Left Hip	$H_l(x), H_l(y)$	Patient
Right Hip	$H_r(x), H_r(y)$	Patient
Left Knee	$K_l(x), K_l(y)$	Patient
Right Knee	$K_r(x), K_r(y)$	Patient
Left Ankle	$A_l(x), A_l(y)$	Patient
Right Ankle	$A_r(x), A_r(y)$	Patient
Left Hand	$HA_l(x), HA_l(y)$	Examiner
Right Hand	$HA_r(x), HA_r(y)$	Examiner

So x and y in the equation represent 2D screen coordinate system. See Table 3 for details.

One action of postural stability may exist multiple values of the straightness at different times. We calculate the minimum/standard deviation/maximum for the upper body, body, and lower body straightness respectively. So there are nine dimensions in total. Besides, the interaction of the examiner pulling the patient's shoulder also plays an important part in postural stability. The backward step and the supporting state are two interactive features. The backward step is the number of the steps back when patients keep postural stability and the steps can be directly counted by the alternations of the left ankle and the right ankle. Supporting means that the examiner uses their hand to support the patient, protecting the patient from falling. This is the signal of the patients losing stability. We use a binary value to represent the supporting state: zero and one. Zero represents no support and one represents that there exists at least one support. The supporting state can be recognized by whether the distance from the shoulder key point of the patient to the hand key point of the examiner is below a threshold. Finally, we get eleven features in total: the minimum/standard deviation/maximum values for three straightness features, one value for the backward step, and one value for the supporting state.

In the final classification stage, eleven features represent eleven dimensions. We feed the data in eleven dimensions to train a classifier. Any classification algorithms can be adopted here. In our postural stability data set, we classify three types: normal, slight, and moderate of MDS-UPDRS.

V. EXPERIMENT

We conduct experiments on the finger tapping and the postural stability. The data set for each action is randomly split into two parts: 60% for training and 40% for testing, and we ensure that different sets have distinguished patients. We hold out 20% training set for validation.

In each experiment, we will report precision, recall, and f1-score in the test set for each class. The micro average is used to evaluate the multi-label performance.

A. FINGER TAPPING

We use MMPose [44] as our pose estimation algorithm. With MMPose [44] toolkit, we adopt Deeppose [45] with ResNet [46] backbone as the basic hand pose algorithm. Deepose is trained on OneHand10K [41] with over 10k RGB hand frames. The output of the model contains 21 hand key

$$\begin{aligned}
\text{US} = 0.5\times \Big| &\frac{(H_l(y)+H_r(y)-E_l(y)-E_r(y))\times(S_l(x)+S_r(x))}{\sqrt{(H_l(x)+H_r(x)-E_l(x)-E_r(x))^2+(H_l(y)+H_r(y)-E_l(y)-E_r(y))^2}}\\
&+\frac{(H_l(x)+H_r(x))\times(E_l(y)+E_r(y))-(E_l(x)+E_r(x))\times(H_l(y)+H_r(y))}{\sqrt{(H_l(x)+H_r(x)-E_l(x)-E_r(x))^2+(H_l(y)+H_r(y)-E_l(y)-E_r(y))^2}}\\
&-\frac{(H_l(x)+H_r(x)-E_l(x)-E_r(x))\times(S_l(y)+S_r(y))}{\sqrt{(H_l(x)+H_r(x)-E_l(x)-E_r(x))^2+(H_l(y)+H_r(y)-E_l(y)-E_r(y))^2}}\Big| \qquad (1)
\end{aligned}$$

$$\begin{aligned}
\text{BS} = 0.5\times \Big| &\frac{(A_l(y)+A_r(y)-S_l(y)-S_r(y))\times(H_l(x)+H_r(x))}{\sqrt{(A_l(x)+A_r(x)-S_l(x)-S_r(x))^2+(A_l(y)+A_r(y)-S_l(y)-S_r(y))^2}}\\
&+\frac{(A_l(x)+A_r(x))\times(S_l(y)+S_r(y))-(S_l(x)+S_r(x))\times(A_l(y)+A_r(y))}{\sqrt{(A_l(x)+A_r(x)-S_l(x)-S_r(x))^2+(A_l(y)+A_r(y)-S_l(y)-S_r(y))^2}}\\
&-\frac{(A_l(x)+A_r(x)-S_l(x)-S_r(x))\times(H_l(y)+H_r(y))}{\sqrt{(A_l(x)+A_r(x)-S_l(x)-S_r(x))^2+(A_l(y)+A_r(y)-S_l(y)-S_r(y))^2}}\Big| \qquad (2)
\end{aligned}$$

$$\begin{aligned}
\text{LS} = 0.5\times \Big| &\frac{(A_l(y)+A_r(y)-H_l(y)-H_r(y))\times(K_l(x)+K_r(x))}{\sqrt{(A_l(x)+A_r(x)-H_l(x)-H_r(x))^2+(A_l(y)+A_r(y)-H_l(y)-H_r(y))^2}}\\
&+\frac{(A_l(x)+A_r(x))\times(H_l(y)+H_r(y))-(H_l(x)+H_r(x))\times(A_l(y)+A_r(y))}{\sqrt{(A_l(x)+A_r(x)-H_l(x)-H_r(x))^2+(A_l(y)+A_r(y)-H_l(y)-H_r(y))^2}}\\
&-\frac{(A_l(x)+A_r(x)-H_l(x)-H_r(x))\times(K_l(y)+K_r(y))}{\sqrt{(A_l(x)+A_r(x)-H_l(x)-H_r(x))^2+(A_l(y)+A_r(y)-H_l(y)-H_r(y))^2}}\Big| \qquad (3)
\end{aligned}$$

TABLE 4. Experiment results on the left finger tapping in test set. Precision, recall, and f1-score are reported here. Type 0, type 1, type 2, and type 3 represent the normal, slight, mild, and moderate severity of MDS-UPDRS, respectively. On this tet set, the corresponding number randomly picked from the whole set is 26, 60, 54, and 8. We adopt DNN as the final classification method.

Type	Precision	Recall	F1-score
0	0.89	0.92	0.91
1	0.90	0.88	0.89
2	0.88	0.91	0.89
3	0.83	0.62	0.71
Micro Average	0.88	0.89	0.88

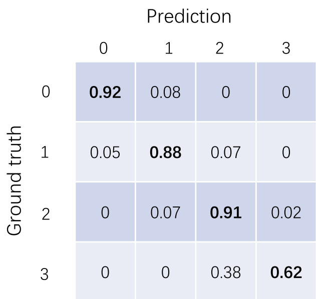


FIGURE 5. Confusion matrix of the left finger tapping in the test set. Type 0, type 1, type 2, and type 3 represent the normal, slight, mild, and moderate severity of MDS-UPDRS, respectively.

TABLE 5. Experiment results on the right finger tapping in test set. Precision, recall, and f1-score are reported here. Type 0, type 1, type 2, and type 3 represent the normal, slight, mild, and moderate severity of MDS-UPDRS, respectively. On this tet set, the corresponding number randomly picked from the whole set is 17, 47, 37, and 3. We adopt DNN as the final classification method.

Type	Precision	Recall	F1-score
0	0.78	0.82	0.80
1	0.84	0.87	0.85
2	0.89	0.86	0.88
3	1.00	0.33	0.50
Micro Average	0.85	0.85	0.84

points. The percent of correct key points with 20% tolerance (PCK@0.2) of this method is 0.99, which can provide a SOTA result for PD hand pose estimation.

After the pose estimation, tapping rate, tapping frozen times, and tapping amplitude variation are extracted for the inputs of the followed classifier. The methods to extract those three features are described in Section IV-A. Then we use the data with three dimensions to train a classifier with a deep neural network (DNN). Our DNN classifier has three layers and each layer has 8 hidden nodes. We use *adam* learner with learning rate: 0.01 and weight decay: $1e$-6. The number of the training epoch is 200 and the training loss is *Softmax*. Then we test the method on the test set. We report the precision, recall, f1-score, and confusion matrix of the DNN-based classifier. Type 0, type 1, type 2, and type 3 represent the normal, slight, mild, and moderate severity of MDS-UPDRS respectively.

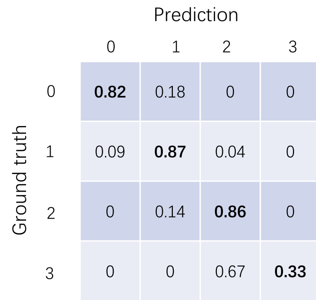


FIGURE 6. Confusion matrix of the right finger tapping in the test set. Type 0, type 1, type 2, and type 3 represent the normal, slight, mild, and moderate severity of MDS-UPDRS, respectively.

TABLE 6. Experiment results on the postural stability in test set. Precision, recall, and f1-score are reported here. Type 0, type 1, and type 3 represent the normal, slight, and moderate severity of MDS-UPDRS respectively. On this tet set, the corresponding number randomly picked from the whole set is 49, 5, and 14. We adopt DNN as the final classification method.

Type	Precision	Recall	F1-score
0	0.85	0.96	0.90
1	1.00	0.60	0.75
3	0.80	0.57	0.67
Micro Average	0.85	0.85	0.84

We test our method on both left and right hands in our experiment, where all experiment setups are shared. The confusion matrices of left and right finger tapping are shown in Figure 5 and Figure 6. The precision, recall, and f1-score are shown in Table 4 and Table 5. The micro averages of left finger tapping in precision, recall, and f1-score are 88%, 89%, and 88%, respectively. The micro averages of right finger tapping in precision, recall, and f1-score are 85%, 85%, and 84%, respectively. Experiment results show that our proposed method can achieve a satisfying result. The difference between the left and right hands is attributed to different data distribution, and the severity is not symmetrical. Besides, we can see that adjacent types are more likely to be confused by two confusion matrices. Because the moderate type occupies the least in our data set, the precision, recall, and f1-score are all relatively lower than others. The train and test loss of DNN classification model are shown in Figure 8 (a) and (b).

B. POSTURAL STABILITY

We also use MMPose [44] as our pose estimation algorithm. In MMPose toolkit, we adopt FasterRCNN [47] to detect persons and HRNet [48] as the pose estimation method. FasterRCNN and HRNet are both trained on COCO dataset [34], a large-scale object recognition and human key points dataset. This dataset covers various scenarios and is suitable for detecting patients and estimating key points in postural stability detection. After the above operations, we get 17 key

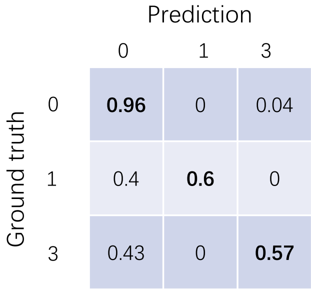


FIGURE 7. **Confusion matrix of the postural stability in the test set. Type 0, type 1, and type 3 represent the normal, slight, and moderate severity of MDS-UPDRS respectively.**

TABLE 7. **Comparison with I3D-based and PoseC3D-based methods. To consistency with our experiment, we report the micro average on the f1-score here.**

Actions	I3D [13]	PoseC3D [29]	Ours
Finger Tapping (Left)	0.41	0.37	0.88
Finger Tapping (Right)	0.34	0.42	0.84
Postural Stability	0.63	0.60	0.84

points for the patient and 17 key points for the examiner in each video frame. The mAP of this method is 74.6%, which can provide a SOTA result for PD body pose estimation.

After the pose estimation, five domain-knowledge-infused features are extracted: upper-body straightness, body straightness, lower-body straightness, backward steps, and supporting status. The methods to extract those three features are described in Section IV-B. We calculate the maximum, the standard deviation, and the minimum number for each straightness. So five domain features consist of eleven dimensions and form eleven dimensions of the data. Then those training data are fed into a DNN classifier same as the finger tapping part. The micro averages of postural stability in precision, recall, and f1-score are 85%, 85%, and 84%, shown in Table 6. The corresponding confusion matrix is shown in Figure 7. Type 0, type 1, and type 3 represent the normal, slight, and moderate severity of MDS-UPDRS, respectively. The result shows that a large portion of the slight and the moderate types are misclassified as the normal severity, showing that domain-knowledge-infused features are still to be explored. The train and test loss of DNN classification model are shown in Figure 8(c).

C. BASELINE METHODS AND ABLATION STUDIES

We compare our results with several baselines: 1) We train a widely-used action recognition method [13] in accessing PD motor severity to form a baseline. 2) We exclude the domain knowledge extraction stage and directly train an action recognition model from extracted key points to show the efficiency of the domain knowledge extraction stage. 3) We test several machine learning methods on the classification stage of our proposed method to show that our proposed method can learn a good representation. The previous two baselines are trained end-to-end, directly taking the original features as the input without an expert-involved domain knowledge refinement stage. The third baseline is based on our proposed method, and we only replace the classification with other machine learning methods.

1) COMPARISON WITH END-TO-END METHODS

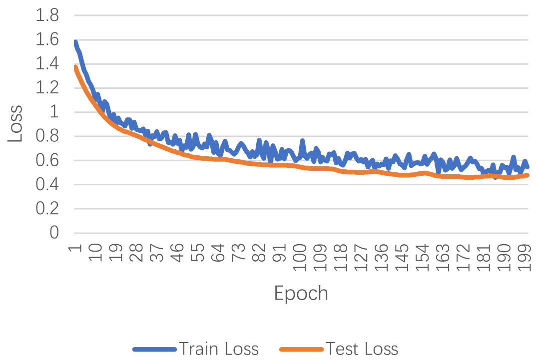
Dadashzadeh *et al.* [13] use I3D [25], [26] algorithm, a classic action recognition method, to measure the PD motor severity in the hand movement and the gait; we compare our method with them. Because their proposed dataset is collected from a different source and not released publicly, we do not directly compare the result with them, and we adopt a similar method, I3D [25], [26].

We use MMAction [49] toolkit to train the I3D-based method. In MMAction, we use a dense sampling strategy for finger tapping and postural stability. We also use data augmentation for finger tapping and only use a flip data augmentation with the probability of 0.5 for the postural stability. We train the 100 epoch and choose the best model for testing. The micro average of the I3D method for the left finger tapping, the right finger tapping, and postural stability is 41%, 34%, and 63%, a large gap from our method. The results are only slightly better than the random ones. The reason is that action recognition like I3D is suitable for coarse actions and not for fine-grained actions like PD motor. Besides, we think training data is too little to train a deep-learning-based method, which requires a large amount of data before achieving a satisfying result. The result is shown in the first column of the Table. 7.

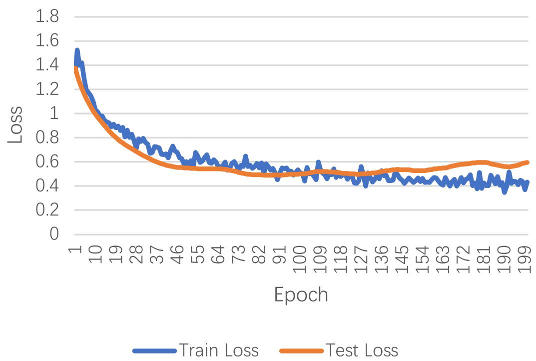
2) DOMAIN KNOWLEDGE EXTRACTION STAGE

To test the efficiency of the domain-knowledge-infused features, we directly train an end-to-end model without this stage. The end-to-end model directly takes the pose key point as the input and outputs the PD motor severity. To achieve this target, we adopt PoseC3D [29] as the end-to-end model. PoseC3D is a 3D-CNN-based approach for skeleton-based action recognition. Instead of directly taking 2D skeletons as the input, PoseC3D takes the key point as the input and forms a 3D heatmap format with three dimensions: x coordinate, y coordinate, and the prediction score. Then data in 3D heatmap format are trained to classify different types of actions. Besides, PoseC3D takes 3D CNN as the backbone, which can capture spatial and temporal features of skeleton sequences.

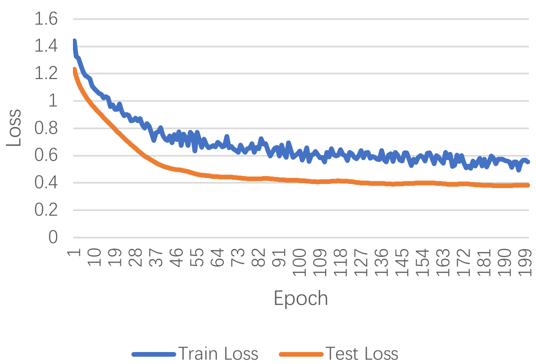
The training stage follows PoseC3D [29]. We first convert the PD key point to the heatmap format, and then the data is used to train a classification model for PD motor severity. We use the uniform sampling strategy for both actions and change the number of sampling clips to 125 for the finger tapping. All frames are uniformly sampled to avoid losing temporal information. The micro average of the PoseC3D method for the left finger tapping, the right finger tapping, and postural stability is 37%, 42%, and 60%. The result on PoseC3D [29] is even worse than I3D. Apart from little training data, the features combined with key points are still sparse



(a) Train and test loss in training the left hand



(b) Train and test loss in training the right hand



(c) Train and test loss in training the postural stability

FIGURE 8. Train and test loss of DNN classification model for the postural stability, the left hand, and the right hand, respectively. The experiments on the train and test loss show that the DNN classification model does not manifest any overfitting signs.

for the network to learn a good representation. After being integrated with domain knowledge extraction, our method boosts the performance. The micro average of our method for the left finger tapping, the right finger tapping, and postural stability is 88%, 84%, and 84%. Apart from adding medical interpretability, domain knowledge extraction has great potential in learning representation from a few samples.

3) CLASSIFICATION MODELS

The classification stage is to classify features into different types of PD motor severity after features are extracted from the domain knowledge extraction stage. Different methods

TABLE 8. Ablations on different classification methods under our unified pipeline. The final stage of our proposed can use various classification methods; we test several methods: SVM, RF, LR, and DNN, which represent support vector machine, random forest, logistic regression, and deep neural network. The micro average of the f1-score here is reported here.

Actions	SVM	RF	LR	DNN
Finger Tapping (Left)	0.80	0.84	0.83	0.88
Finger Tapping (Right)	0.73	0.73	0.74	0.84
Postural Stability	0.75	0.76	0.83	0.84

are adopted in the classification stage of PD motor severity. We test several widely-used methods including support vector machine, random forest, logistic regression, and deep neural network. We use *scikit-learn* [50] toolkit to test those methods. For the support vector machine, we use *SVC* module of *scikit-learn* [50] toolkit. The kernel method is *poly*, the regularization parameter is 10, and the tolerance value for the stopping criterion is $1e$-3. For random forest, we use parameters of *n_estimators*: 9 and *max_depth*: 3; the rest remains the default. For logistic regression, we directly use the default parameter. DNN is a deep neural network with three layers, and each layer has 8 hidden nodes. The detailed architecture and training parameters of DNN are described in Section 8.

The micro average of the f1-score of in support vector machine, random forest, logistic regression, and deep neural network is shown in Table 8. Deep neural network outperforms other machine learning methods, showing the ability to fit data. Machine learning methods (e.g., SVM, RF, LR) have better interpretability while sacrificing algorithm performance, which is a trade-off in practice.

VI. CONCLUSION

Deep-learning-based shows its efficiency in detecting PD severity but requires an extensive training corpus and lacks interpretability, which hinders the wide application of PD severity detection. In this paper, we propose an automatic detection pipeline for PD motor severity in finger tapping and postural stability. This pipeline contains three stages: the pose estimation stage, the domain knowledge extraction stage, and the classification stage. The pose estimation and classification stage are composed of deep learning methods, which provide efficiency in model fitting. In the domain knowledge extraction stage, we propose several domain features by the expert and researchers can adjust those features as they need, which provides interpretability and reduces the need for a large amount of data. To better test our methods, we collect a PD motor dataset from a real hospital. We release the PD motor dataset to the public and we wish to provide a comprehensive testbed for the research community. Experiment shows that our proposed method achieves a satisfying result.

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NING YANG received the master's degree in economics from the University of Macau, Macau, China, in 2017, where he is currently pursuing the Ph.D. degree in electrical and computer engineering. His research interests include artificial intelligence and biomedical engineering.

DE-FENG LIU received the master's degree in neurosurgery from the Capital University of Medical, Beijing, China, in 2019, where he is currently pursuing the Ph.D. degree in neurosurgery. His research interests include diagnosis and treatment of movement disorders.

TAO LIU received the M.Eng. degree from Shanghai Jiao Tong University, in 2021, where he is currently pursuing the Ph.D. degree. His supervisor is Kai Yu. His research interests include multi-modal learning and computer vision.

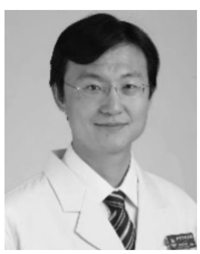
TIANYUAN HAN received the bachelor's degree. He is currently pursuing the master's degree with Shanghai Jiaotong University, China. His research interest includes human pose recognition in video scenarios.

PINGYUE ZHANG received the B.E. degree from Shanghai Jiaotong University, in 2021, where he is currently pursuing the M.E. degree. His research interests include machine learning, self-supervised learning, and rich audio analysis.

XUENAN XU received the B.S. degree from Shanghai Jiao Tong University, in 2019, where he is currently pursuing the Ph.D. degree. His supervisors are Kai Yu and Mengyue Wu. His research interests include detection and classification of acoustic scenes and events, the interaction between audio signal processing, and natural language processing.

SIYU LOU received the B.Eng. degree from Shanghai Jiao Tong University, in 2015, and the dual M.Eng. degree from the Institut Polytechnique de Grenoble and Shanghai Jiao Tong University, in 2016 and 2018, respectively. She is currently working as a Research Assistant at Shanghai Jiao Tong University. Her research interests include multi-modal learning and interpretable neural networks for audio classification.

HUAN-GUANG LIU received the master's and Ph.D. degrees in neurosurgery from the Capital University of Medical, Beijing, China, in 2010 and 2013, respectively. He is engaged in clinical and scientific research in functional neurosurgery.



AN-CHAO YANG received the master's and Ph.D. degrees in neurosurgery from the Capital University of Medical, Beijing, China, in 2004 and 2011, respectively. He is engaged in clinical and scientific research in functional neurosurgery.

CHENG DONG received the B.Sc. degree from Nanjing University, Nanjing, China, in 2008, and the M.Sc. and Ph.D. degrees in electrical and electronics engineering from the University of Macau, Macau, China, in 2012 and 2017, respectively. Since 2019, he has been a Lecturer and a Postdoctoral Fellow with the School of Intelligent Systems Science and Engineering, Jinan University, China. His current research interests include digital signal processing, electronic automation in the digital microfluidic systems, and molecular biology for disease detection.

MANG I. VAI (Senior Member, IEEE) received the Ph.D. degree in electrical and electronics engineering from the University of Macau, Macau, China, in 2002. Since 1984, he has been involved in research in the areas of digital signal processing and embedded systems. He is currently an Associate Professor with the Department of Electrical and Computer Engineering, Faculty of Science and Technology, and the State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau.

SIO HANG PUN received the master's degree in computer and electrical program from the University of Porto, Porto, Portugal, in 1999, and the Ph.D. degree in electrical and electronics engineering from the University of Macau, Macau, China, in 2011. Since 2000, he has been performed research in the area of biomedical engineering. He is currently an Associate Professor with the State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau. His research interests include bioelectronic circuits, intra-body communications, and bioelectromagnetism.

JIAN-GUO ZHANG received the master's degree in anatomy from the Jiamusi Medical College, Jiamusi, China, in 1990, and the Ph.D. degree in surgery from Tongji Medical University, Hubei, China, in 1996. He is engaged in clinical and scientific research in functional neurosurgery.